

PODGORICA GRAD, Montenegro

WMO: 134630

Lat: 42.433N

Lon: 19.283E

Elev: 50

StdP: 100.73

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.9	-1.4	-13.2	1.2	2.0	-10.9	1.5	3.1	13.4	3.9	12.1	3.4	3.2	0	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	37.0	21.9	35.3	21.5	34.0	21.2	23.4	32.6	22.7	32.1	22.1	31.3	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	15.6	27.1	20.0	14.7	26.1	19.1	13.9	25.6	70.0	32.7	67.2	32.2	64.7	31.3	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.6	7.1	-5.5	39.0	1.7	1.9	-6.7	40.4	-7.7	41.5	-8.7	42.5	-9.9	43.9		
(5)			-6.8	25.1	1.8	1.5	-8.1	26.2	-9.2	27.0	-10.2	27.8	-11.5	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	5.9	7.4	10.8	15.0	19.6	24.3	27.3	27.3	21.8	16.8	11.5	7.1		
	DBStd	8.15	3.15	3.35	3.27	3.14	3.07	3.48	2.74	2.91	3.16	3.12	3.40	3.21		
	HDD10.0	371	131	84	29	3	0	0	0	0	0	1	24	100		
	HDD18.3	1683	386	307	232	105	23	2	0	0	6	66	207	349		
	CDD10.0	2661	3	10	55	154	296	429	537	536	354	211	67	10		
	CDD18.3	932	0	0	0	6	61	181	279	277	110	18	0	0		
	CDH23.3	10128	0	1	2	70	507	1916	3366	3269	887	111	1	0		
(13)	CDH26.7	4763	0	0	0	12	127	844	1751	1713	305	11	0	0		
(14)	Wind	WSAvg	2.5	2.5	2.7	3.0	2.5	2.5	2.8	2.6	2.4	2.1	2.1	2.5		
(15) Precipitation	PrecAvg	1707	170	158	143	147	90	57	46	66	135	178	242	217		
	PrecMax	2359	381	385	349	340	230	162	140	274	390	523	639	413		
	PrecMin	987	0	0	6	14	0	8	0	5	0	0	20	36		
	PrecStd	311	114	99	81	75	55	40	30	61	105	124	122	107		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.5	18.8	23.1	28.2	31.2	36.4	38.6	39.2	34.9	28.0	22.2	16.9		
		MCWB	10.9	11.5	13.2	16.9	19.6	22.4	22.0	22.2	20.6	18.6	15.7	13.0		
	2%	DB	13.9	16.1	20.4	25.1	29.8	34.8	37.0	37.4	32.0	26.0	20.1	15.1		
		MCWB	10.2	10.6	12.5	15.4	19.1	22.1	22.0	21.6	19.7	17.8	14.8	12.1		
	5%	DB	12.2	14.8	18.8	23.1	28.0	33.0	35.2	35.8	30.0	24.2	18.3	13.9		
		MCWB	9.0	9.9	11.8	14.5	18.4	21.4	21.6	21.3	19.6	17.1	14.0	10.9		
	10%	DB	10.9	13.0	16.9	21.2	26.2	31.2	33.9	34.1	28.1	22.8	16.9	12.2		
		MCWB	8.2	8.8	10.9	13.8	17.7	20.8	21.2	21.0	19.0	16.5	13.5	9.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	13.4	15.9	17.9	21.1	24.3	24.6	24.6	22.3	20.5	17.4	14.5		
		MCDB	14.4	16.0	18.8	26.0	28.9	33.0	33.1	32.9	28.8	25.1	19.9	15.5		
	2%	WB	11.2	12.0	13.9	16.6	19.8	23.1	23.4	23.3	21.4	19.3	16.1	13.1		
		MCDB	12.9	14.9	18.0	23.3	27.6	32.5	32.8	32.5	28.2	23.5	18.8	14.4		
	5%	WB	10.1	10.9	12.8	15.5	19.0	22.2	22.7	22.6	20.7	18.3	15.1	11.8		
		MCDB	11.2	13.5	16.8	21.1	26.4	31.2	32.4	31.9	27.4	22.1	17.5	13.2		
	10%	WB	9.0	9.9	11.8	14.7	18.2	21.3	22.0	21.8	19.9	17.4	14.0	10.4		
		MCDB	10.1	11.9	15.4	19.6	25.0	29.8	31.6	31.2	26.2	21.0	16.1	11.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.8	9.3	10.1	10.5	11.1	11.7	11.9	10.6	10.1	8.9	8.1		
		MCDBR	10.2	11.9	12.6	13.4	13.2	12.8	13.5	14.0	12.9	12.5	10.9	9.8		
		MCWBR	6.8	7.2	6.7	6.2	5.7	5.1	4.9	5.1	5.2	6.2	6.3	6.5		
		MCDBR	7.6	9.2	10.4	11.7	12.0	12.2	12.4	12.7	11.4	10.1	8.7	6.7		
	5% WB	MCWBR	5.7	6.1	6.2	5.8	5.5	5.2	5.1	5.3	5.3	5.5	5.5	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.343	0.371	0.424	0.484	0.473	0.453	0.454	0.466	0.441	0.405	0.372	0.342		
	taud		2.362	2.254	2.137	2.007	2.101	2.215	2.211	2.171	2.222	2.298	2.337	2.374		
	Ebn at Noon		783	815	810	788	809	826	820	796	786	768	738	747		
	Edh at Noon		83	108	137	168	157	140	139	141	125	103	85	76		
(44) All-Sky Solar Radiation	RadAvg		1.62	2.34	3.47	4.54	5.65	6.57	6.80	6.08	4.33	3.00	1.88	1.45		
	RadStd		0.20	0.35	0.50	0.52	0.47	0.46	0.40	0.42	0.37	0.32	0.21	0.28		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.58	N/A	-1.36	N/A	N/A	N/A	N/A	-141	N/A	N/A
(47)	Regional (27 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	-103	N/A	N/A

Nomenclature: See separate page